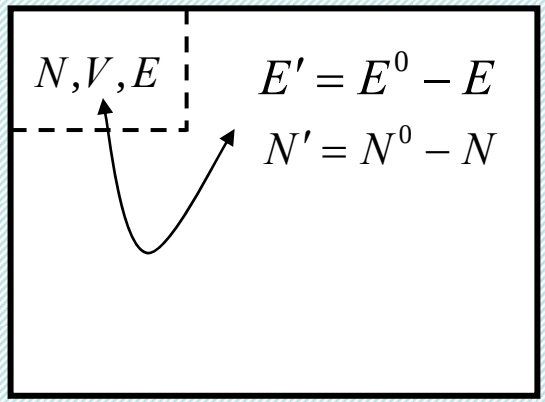
§4-4 巨正则系综（Grand-Canonical Ensemble）

- 由大量具有确定的化学势 μ、体积 V 和温度 T 的体系所组成的系综称为巨正则系综。
- 考虑一个具有确定的化学势 μ、体积 V 和温度 T 的系统 A，与一个大热源和粒子源 A' 彼此接触、处于平衡状态，并构成一个大的孤立体系。

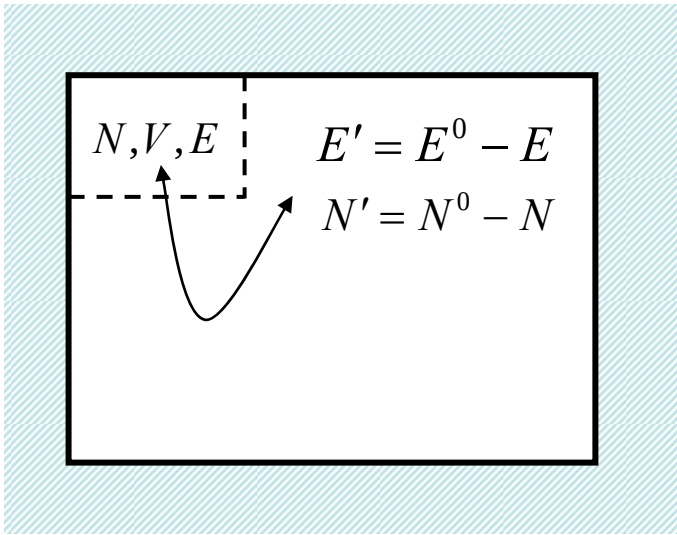
$$E^0 = E + E' \qquad E' >> E$$

$$N^0 = N + N' \qquad N' >> N$$



- 根据平衡态孤立体系的性质，体系 A 处于能量为 E，粒子数为 N 的可实现微观态的几率为：

$$\rho(E,N)=\frac{\Omega(E,N)\Omega'(E',N')}{\Omega^0(E^0,N^0)}$$



- 而体系 A 处于能量为 E，粒子数为 N 的某一微观态的几率：

$$\rho_r=\frac{\Omega'(E',N')}{\Omega^0(E^0,N^0)}$$

- 利用 $E'>>E \quad N'>>N$ 的条件，将 E' N'的自然对数在 E^0 N^0附近展开：

$$\Rightarrow \quad \ln\Omega'(E',N') = \ln\Omega'(E^0 - E_r, N^0 - N_r)$$

$$\approx \ln\Omega'(E^0,N^0) - \left[\frac{\partial \ln\Omega'}{\partial E'}\right]E_r - \left[\frac{\partial \ln\Omega'}{\partial N'}\right]N_r$$

$$= \ln\Omega'(E^0,N^0) - \beta E_r - \alpha N_r$$

- 其中：$\beta = \left[\frac{\partial \ln\Omega'}{\partial E'}\right]_{E'=E^0}$ ； $\alpha = \left[\frac{\partial \ln\Omega'}{\partial N'}\right]_{N'=N^0}$

$$\Rightarrow \quad \Omega'(E',N') = \Omega'(E^0,E^0)\cdot e^{-\beta E_r - \alpha N_r}$$

$$\Rightarrow \quad \rho_r = \frac{\Omega'(E',N')}{\Omega^0(E^0,N^0)} = \frac{\Omega'(E^0,N^0)}{\Omega^0(E^0,N^0)}\cdot e^{-\beta E_r - \alpha N_r} = C\cdot e^{-\beta E_r - \alpha N_r}$$



- 利用归一化条件：$\sum_r \rho_r = 1$

$$\Rightarrow \quad \sum_r C e^{-\beta E_r - \alpha N_r} = 1$$

$$\Rightarrow \quad C = \frac{1}{\sum_r e^{-\beta E_r - \alpha N_r}} = \frac{1}{\Xi}$$

- $\Xi = \sum_r e^{-\beta E_r - \alpha N_r}$ 称为巨正则系综的配分函数。

- $\rho_r = \frac{1}{\Xi} e^{-\beta E_r - \alpha N_r}$ 则为系统 A 在化学势为 μ，温度为 T 的条件下，处于能量为 E 和粒子数为 N 的某微观态的概率。

- 对于能量为 E 和粒子数为 N 的所有微观态，有：

$$\sum_{E,N} \rho(E,N) = 1$$

$$\Rightarrow \quad \sum_{E,N} C\Omega(E,N) e^{-\beta E - \alpha N} = 1$$

$$\Rightarrow \quad C = \frac{1}{\sum_{E,N} \Omega(E,N) e^{-\beta E - \alpha N}} = \frac{1}{\Xi}$$

$$\Rightarrow \quad \rho(E,N) = \frac{1}{\Xi} \Omega(E,N) e^{-\beta E - \alpha N}$$



- 巨正则配分函数还可以表示为：

$$\Xi(\alpha,V,\beta)=\sum_{E,N}\Omega(E,N)e^{-\beta E-\alpha N}$$

$$=\sum_{N}\left[e^{-\alpha N}\sum_{E}\Omega(E)e^{-\beta E}\right]$$

$$=\sum_{N}\left(e^{-\alpha N}Z_N(V,\beta)\right)$$

- 定义逸度： $f=e^{-\alpha}=e^{\beta\mu}$
- 它代表了体系在所处的状态下，分子逃逸的难易程度，或称逸散能力

系综的等同性

$$\Rightarrow \quad \Xi(\alpha,V,\beta)=\sum_N f^N Z(V,\beta)$$

- 通常写成： $\Xi(\mu,V,T)=\sum_N f^N Z(V,T)$
- 如果系统的粒子数很大，当系统达到平衡时，体系的粒子数涨落其实很小，体系的微观态的粒子数目几乎都在平均粒子数附近。
- 在此条件下，巨正则系综和处于平均粒子数下的正则系综其实是等同的。
- 这也称为系综的热力学等同性。



例 4.3 吸附现象

设吸附表面有 N_0 个吸附中心，每个吸附中心可吸附一个气体分子。被吸附的气体分子能量为 $-\varepsilon_0$，求达到平衡时吸附率 $\theta = N/N_0$ 与气体温度和压强的关系。

解：将气体看作热源和粒子源。被吸附的分子系统可看作与气体（源）交换粒子和能量的系统，遵从巨正则分布。求 θ 的关键是求出 N

当有 N 个分子被吸附时，系统的能量为“$-N\varepsilon_0$”，系统的巨配分函数为：

$$\Xi = \sum_{N=0}^{N_0}\sum_{r} e^{-\alpha N-\beta E_r} = \sum_{N=0}^{N_0} e^{\beta\mu N} Z_N(T,A)$$



式中 $Z_N(T, A)$ 是吸附面上有N 个分子的正则分布配分函数：

$$Z_N(T,A)=\sum_r e^{\beta\varepsilon_0 N}=e^{\beta\varepsilon_0 N}\frac{N_0!}{N!(N_0-N)!}$$

上式考虑到 N 个分子在 N_0 个吸附中心上的组合，即正则系综具有的微观状态数。

$$\Rightarrow\quad \Xi=\sum_{N=0}^{N_0}e^{\beta\mu N}Z_N(T,A)=\sum_{N=0}^{N_0}e^{\beta(\mu+\varepsilon_0)N}\frac{N_0!}{N!(N_0-N)!}$$

$$=\left[1+e^{\beta(\mu+\varepsilon_0)}\right]^{N_0}$$

二项式定理：$(a+b)^n=C_n^0a^nb^0+C_n^1a^{n-1}b^1+\cdots+C_n^na^0b^n$



被吸附分子的平均数：

$$\Xi = \left[1 + e^{\beta(\mu+\varepsilon_0)}\right]^{N_0}$$

$$\overline{N} = -\frac{\partial}{\partial\alpha}\ln\Xi = kT\frac{\partial}{\partial\mu}\ln\Xi = \frac{N_0}{1+e^{-\beta(\mu+\varepsilon_0)}}$$

$$\Rightarrow \quad \theta = \frac{\overline{N}}{N_0} = \frac{1}{1+e^{-\beta(\mu+\varepsilon_0)}}$$

达到吸附平衡时，吸附系统与气体源的化学势和温度应相等。

如果气体是单原子分子理想气体：

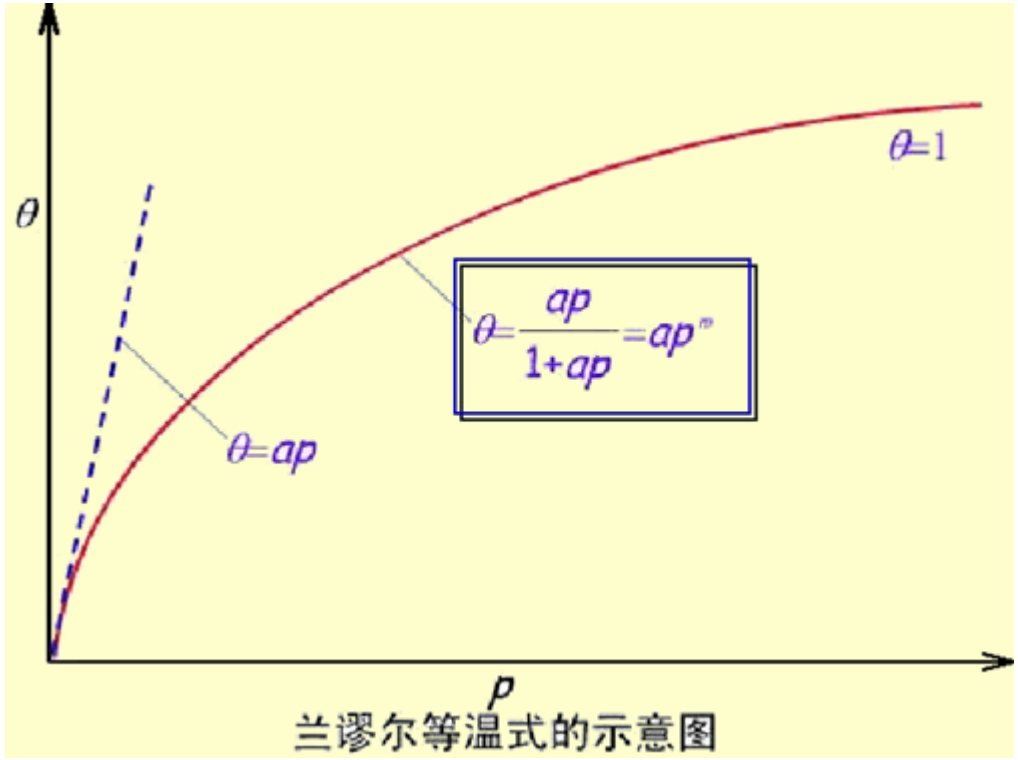
$$e^{\alpha} = \frac{Z}{N} = \frac{V}{N}\left(\frac{2\pi m k_B T}{h^2}\right)^{3/2} = \frac{k_B T}{p}\left(\frac{2\pi m k_B T}{h^2}\right)^{3/2} = e^{-\beta\mu}$$



Langmuir 等温吸附公式

$$\theta = \frac{1}{1+\frac{k_B T}{p}\left(\frac{2\pi m k_B T}{h^2}\right)^{3/2} e^{-\frac{\varepsilon_0}{k_B T}}}$$

$$\theta = \frac{ap}{1+ap} = \frac{1}{1+\frac{1}{ap}}$$



兰谬尔等温式的示意图

上式称为 Langmuir 等温吸附公式，式中 θ 为吸附表面覆盖率，a 为吸附系数，后者代表了固体表面吸附气体的能力。

四类常用系综一览

	微正则系综	正则系综	巨正则系综	*T-P*系综
体系的特点	孤立体系	闭口体系	开口体系	闭口体系
约束条件	E, V, N	β, V, N	β, V, α	β, π, N
几率分布	$P_r = C$	$P_r = Ce^{-\beta E_r}$	$P_r = Ce^{-\alpha N_r - \beta E_r}$	$P_r = Ce^{-\beta E_r - \pi V_r}$
配分函数	$\Omega(E,V,N) = \frac{1}{C}$	$Z(T,V,N) = \sum_r \exp(-\frac{E_r}{kT})$	$\Xi(\alpha,V,T) = \sum_{N=0}^{\infty} Z_N(V,T)e^{\frac{\mu N}{kT}}$	$\Delta(T,P,N) = \sum Z_V(T,N)e^{-\frac{PV}{kT}}$
热力学关系	$S(E,V,N) = k\ln\Omega(E,V,N)$	$F(T,V,N) = -kT\ln Z(T,V,N)$	$J(\alpha,V,T) = -kT\ln\Xi(f,V,T)$	$G(T,P,N) = -kT\ln\Delta(T,P,N)$
说明：	$\beta = \frac{1}{kT}$,	$\alpha = -\frac{\mu}{kT}$,	$\pi = \frac{P}{kT}$	

§4-5 系综物理量的涨落

- 统计物理中宏观量是对应的微观量的统计平均值，因此宏观性质会出现围绕平均值的涨落。
- 宏观量围绕平均值的涨落指的是宏观量的瞬时值与它的平均值的偏差。
- 平均值的偏差定义为：

$$\Delta A = A - \overline{A} \qquad \Rightarrow \quad \overline{(\Delta A)} = 0$$

- 重新定义平均平方差：

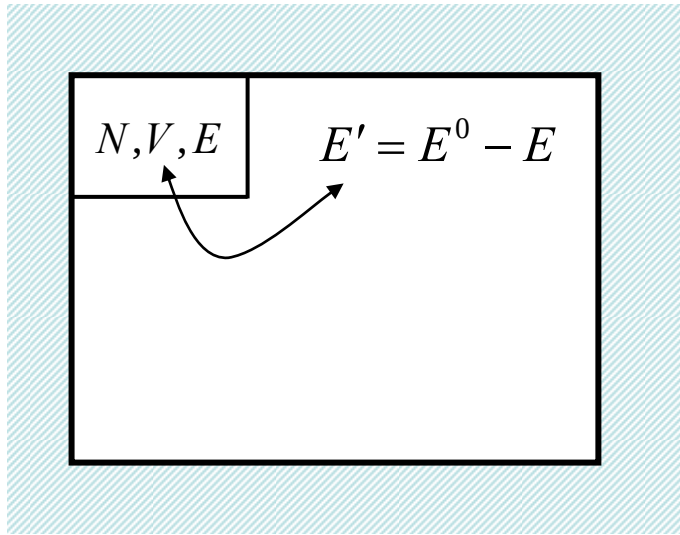
$$(\Delta A)^2 = (A - \overline{A})^2$$

$$\Rightarrow \quad \overline{(\Delta A)^2} = \overline{(A - \overline{A})^2} = \overline{(A^2 - 2A\overline{A} + \overline{A}^2)} = \overline{A^2} - \overline{A}^2$$



- 相对涨落则定义为：$\dfrac{\overline{(\Delta A)^2}}{\overline{A}^2}=\dfrac{\overline{A^2}-\overline{A}^2}{\overline{A}^2}=\dfrac{\overline{A^2}}{\overline{A}^2}-1$

- 现考虑一正则系综：



$$\rho(E)=\frac{1}{Z}\Omega(E)e^{-\beta E}$$

$$\Rightarrow\quad \overline{E}=\frac{\sum_E E\Omega(E)e^{-\beta E}}{\sum_E \Omega(E)e^{-\beta E}}$$

$$\Rightarrow\quad \frac{\partial\overline{E}}{\partial\beta}=-\frac{\sum_E E^2\Omega(E)e^{-\beta E}}{\sum_E \Omega(E)e^{-\beta E}}+\frac{\left(\sum_E E\Omega(E)e^{-\beta E}\right)^2}{\left(\sum_E \Omega(E)e^{-\beta E}\right)^2}=-\overline{E^2}+\overline{E}^2$$

$$\Rightarrow \quad \overline{(\Delta E)^2} = \overline{E^2} - \overline{E}^2 = -\frac{\partial \overline{E}}{\partial \beta}$$

$$= k_B T^2 \frac{\partial \overline{E}}{\partial T}$$

- 对于单原子理想气体：

$$\overline{E} = \frac{3N}{2} k_B T$$

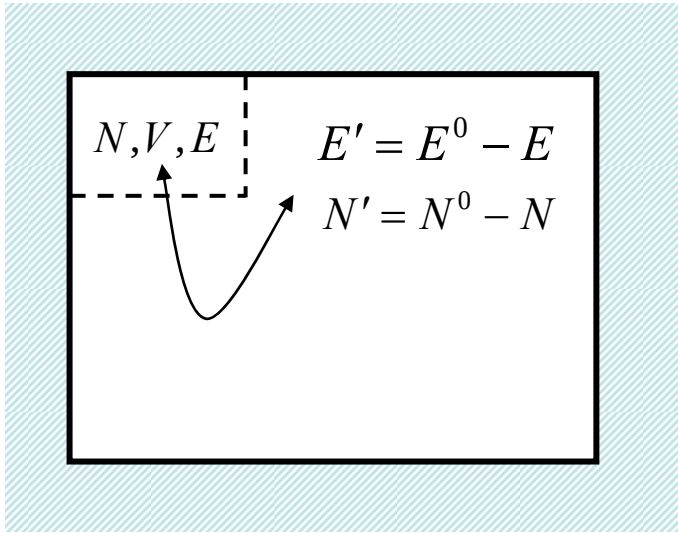
$$\Rightarrow \quad \frac{\overline{(\Delta E)^2}}{\overline{E}^2} = \frac{k_B T^2 \cdot \frac{3N}{2} k_B}{\left(\frac{3N}{2} k_B T\right)^2} = \frac{2}{3N}$$

- 能量的相对涨落与粒子数的倒数成正比。



- 如果考虑巨正则系综的粒子数涨落：

$$\rho(E,N)=\frac{\Omega(E,N)e^{-\beta E-\alpha N}}{\sum_{E,N}\Omega(E,N)e^{-\beta E-\alpha N}}$$



$$\Rightarrow\quad \overline{N}=\frac{\sum_{E,N}N\Omega(E,N)e^{-\beta E}e^{-\alpha N}}{\sum_{E,N}\Omega(E,N)e^{-\beta E}e^{-\alpha N}}$$

$$\frac{\partial\overline{N}}{\partial\alpha}=-\frac{\sum_{E,N}N^2\Omega(E,N)e^{-\beta E}e^{-\alpha N}}{\sum_{E,N}\Omega(E,N)e^{-\beta E}e^{-\alpha N}}+\frac{\left(\sum_{E,N}N\Omega(E,N)e^{-\beta E}e^{-\alpha N}\right)^2}{\left(\sum_{E,N}\Omega(E,N)e^{-\beta E}e^{-\alpha N}\right)^2}=-\overline{N^2}+\overline{N}^2$$

$$\Rightarrow \quad \overline{(\Delta N)^2} = \overline{N^2} - \overline{N}^2 = -\frac{\partial \overline{N}}{\partial \alpha}$$

$$= -\left(\frac{\partial \overline{N}}{\partial \alpha}\right)_{\beta,V} = k_B T \left(\frac{\partial \overline{N}}{\partial \mu}\right)_{T,V}$$

- 可以证明相对涨落有：

$$\left(\frac{\partial \mu}{\partial v}\right)_T = v\left(\frac{\partial p}{\partial v}\right)_T \quad ; \quad v = \frac{V}{\overline{N}}$$

$$\frac{\overline{(\Delta N)^2}}{\overline{N}^2} = \frac{k_B T}{\overline{N}^2}\left(\frac{\partial \overline{N}}{\partial \mu}\right)_{T,V} \propto \frac{1}{\overline{N}}$$

- 巨正则系综粒子数的涨落与平均粒子数的倒数成正比。



Why Is the Sky Blue ?



Tyndall Effect or Rayleigh Scattering ?

- The first steps towards correctly explaining the colour of the sky were taken by John Tyndall in 1859. He discovered that when light passes through a clear fluid holding small particles in suspension, the shorter blue wavelengths are scattered more strongly than the red.
- This is most correctly called the Tyndall effect, but it is more commonly known to physicists as Rayleigh Scattering - after Lord Rayleigh (John William Strutt), who studied it in more detail a few years later. He showed that the amount of light scattered is inversely proportional to the fourth power of wavelength for sufficiently small particles. It follows that blue light is scattered more than red light by a factor of $(700/400)^4$ ~ =10.



Dust or Molecule ?

- Tyndall and Rayleigh thought that the blue color of the sky must be due to small particles of dust and droplets of water vapor in the atmosphere. Even today, people sometimes incorrectly say that this is the case.
- Later scientists realized that if this was true, there would be more variation of sky color with humidity or haze conditions than was actually observed, so they supposed correctly that the molecules of oxygen and nitrogen in the air are sufficient to account for the scattering. The case was finally settled by Einstein in 1911, who calculated the detailed formula for the scattering of light from molecules; and this was found to be in agreement with experiment.



Why Not Violet ?

- The spectrum of light emission from the sun is not constant at all wavelengths, and additionally is absorbed by the high atmosphere, so there is less violet in the light. Our eyes are also less sensitive to violet. That's part of the answer.

- The rest of the answer to this puzzle lies in the way our vision works. We have three types of color receptors, or cones, in our retina. They are called red, blue and green because they respond most strongly to light at those wavelengths. As they are stimulated in different proportions, our visual system constructs the colors we see.

http://www.desy.de/user/projects/Physics/General/BlueSky/blue_sky.html



光的散射

- 根据瑞利散射公式，当单位面积单位强度的光波射入一个小体积 V 后，在和入射光垂直的方向上，每单位立体角所散射的光强为：

$$I = \frac{2\pi^2 V^2}{\lambda^4}\left(\frac{\Delta\mu}{\mu}\right)^2$$

其中 μ 为介质折射率，Δ μ 为折射率的偏差。

- 而根据洛伦兹折射公式：

$$\frac{1}{\rho}\frac{\mu^2-1}{\mu^2+2} = Const.$$

其中 ρ 为介质的密度。



$$\Rightarrow \quad \frac{\Delta\rho}{\rho} = \frac{6\mu^2}{(\mu^2+2)(\mu^2-1)}\frac{\Delta\mu}{\mu}$$

$$\Rightarrow \quad I = \frac{\pi^2 V^2(\mu^2+2)^2(\mu^2-1)^2}{18\lambda^4\mu^4}\left(\frac{\Delta\rho}{\rho}\right)^2$$

- 而根据体积涨落公式可得：

$$\overline{\left(\frac{\Delta\rho}{\rho}\right)^2} = \overline{\left(-\frac{\Delta V}{V}\right)^2} = -\frac{k_B T}{V^2}\left(\frac{\partial V}{\partial p}\right)_T$$

$$\Rightarrow \quad I = \frac{\pi^2 k_B T(\mu^2+2)^2(\mu^2-1)^2}{18\lambda^4\mu^4}\left(\frac{\partial V}{\partial p}\right)_T$$



- 若介质是理想气体：

$$\mu \approx 1 \quad ; \quad \left(\frac{\partial V}{\partial p}\right)_T = -\frac{V}{p}$$

$$\Rightarrow \quad I = \frac{\pi^2(\mu^2-1)^2}{2\lambda^4}\frac{k_B T V}{p}$$

- 表明理想气体的散射光强与波长的四次方成反比。
- 另外，如果介质趋近于临界点：$\left(\frac{\partial V}{\partial p}\right)_T \to \infty$
- 各种波长的光均被散射，液体呈乳白色，称为临界乳光现象。



6月7日课外习题

1、汪志诚习题 9.13

2、汪志诚习题 9.15

